



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

CYBER SECURITY [IT-3252]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Describe the steps in detail to be followed by the organization to perform the security risk assessment. (5)
 - A)
 - B) Determine and explain in detail with examples, the scope of the unified threat management appliance for attacks, internet threats, and protection mechanisms with respect to SMTP protocol. (5)
- 2) Identify and explain in detail the Security controls and mechanisms which can be used to stop the Weaponization and Exploitation stage of cyber kill chain. (5)
 - A)
 - B) Identify and explain in detail SQL injection attacks and discuss any one technique to detect and mitigate. (5)
- 3) Consider you are an attacker who wants to perform an attack on a system. As part of it identify the different types of Port Scans with proper explanation (5)
 - A)
 - B) Investigate and demonstrate the working of the tool with examples ,which is used during the reconnaissance phase which gathers information that is publicly available. (5)
- 4) Explore the Architecture and working of the Distributed Intrusion Detection system with the explanation. (5)
 - A)
 - B) Outline with proper explanation the snort rule format with different types of actions and their descriptions. (5)
- 5) Examine and explain in detail with the help of the diagram the iptables packet flow. (5)
 - A)
 - B) Identify and explain in detail the default Tables and Chains in Iptables (5)

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